

Portland State University

DEPARTMENT OF
ELECTRICAL AND COMPUTER ENGINEERING

ECE332

LAB 4: PATCH ANTENNA

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Term: Spring 2026

Date: May 14, 2026

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1 Introduction

This lab builds and characterizes a half-wave rectangular patch antenna on single-sided copper-clad FR4, tuned for the 2.4 GHz WiFi band ($f_c \approx 2.443$ GHz, the geometric mean of the 2.400–2.487 GHz channel range). The patch is fed at the edge by a $50\ \Omega$ microstrip transmission line made from copper tape, and a side-launch SMA connector ties the transmission line to the VNA. The patch length L sets the resonant frequency through the half-wavelength-in-dielectric condition $f_c = c/(2L\sqrt{\epsilon_r})$. The width W controls the input impedance and the radiation pattern, and the inset distance R tunes the feed-point impedance down toward $50\ \Omega$.

After construction we measure S_{11} and S_{22} on a VNA to verify the resonance lands inside the WiFi band, trim the patch length if needed, then cut an inset feed of length R to drop the input impedance to $50\ \Omega$. The second half of the lab uses both antennas at once to measure S_{21} and S_{12} versus orientation angle, producing an azimuth pattern of the radiation lobe. A cross-polarization test (90° rotation of one antenna) verifies that patch antennas are linearly polarized in the far field.

2 Antenna Design Calculations

Patch length L versus ϵ_r

The center frequency of the patch is set by the half-wavelength-in-dielectric condition,

$$f_c \approx \frac{c}{2L\sqrt{\epsilon_r}} \implies L \approx \frac{c}{2f_c\sqrt{\epsilon_r}}.$$

With $c = 2.998 \times 10^8$ m/s and $f_c = \sqrt{f_1 f_2} = \sqrt{2.400 \cdot 2.487} \approx 2.443$ GHz (the geometric mean of the WiFi band), sweeping ϵ_r across the FR4 spec range gives:

ϵ_r	L (mm)
4.0	30.806
4.1	30.428
4.2	30.063
4.3	29.711
4.4	29.372

About the spread: a 10% sweep in ϵ_r (4.0 to 4.4) only moves L by about 1.43 mm, which is roughly 4.7% of L . The length scales as $1/\sqrt{\epsilon_r}$, so the sensitivity is half the relative error in ϵ_r . For example a ± 0.1 swing around $\epsilon_r = 4.2$ shifts L by about ± 0.36 mm, which is right at the edge of what an Xacto knife can resolve. So basically picking the wrong ϵ_r inside the FR4 spec range won't kill the resonance, it just nudges f_c by a few tens of MHz and we trim by hand on the VNA anyway.

Design choice: use the middle value $L = 30$ mm for the initial cut, which corresponds to $\epsilon_r \approx 4.2$.

Patch width W

The constraint from the lab is $W \geq L$, with larger W dropping the input impedance and broadening the pattern. A square patch ($W = L \approx 30$ mm) sits around $300\ \Omega$. We picked $W = 67$ mm, which

is the short edge of the FR4 board minus 2 mm of clearance margin. About $2.2\times$ wider than L , which drops the edge-fed impedance well below the square-patch $300\ \Omega$ value and uses the full board width for the radiating aperture. W runs along the short edge of the board so the transmission line and SMA connector fit along the long edge.

50 Ω transmission line width

The 50 Ω microstrip trace width was computed using the Digi-Key PCB Trace Impedance Calculator¹ with the measured substrate parameters:

ε_r	4.2
Copper thickness t	0.04 mm
Substrate height h	1.40 mm
Target Z_0	50 Ω
Trace width w	2.632 mm

We used $w = 2.632$ mm for the transmission line cut. The slightly thinner-than-spec substrate ($h = 1.40$ mm measured vs 1.6 mm nominal) and the copper-tape thickness ($t = 0.04$ mm) were both factored into the calculator instead of using the textbook approximation.

Wavelength and far-field boundary

The free-space wavelength at $f_c = 2.443$ GHz is

$$\lambda_0 = \frac{c}{f_c} = \frac{2.998 \times 10^8}{2.443 \times 10^9} \approx 12.28 \text{ cm.}$$

For an aperture antenna with largest dimension D , the far-field (Fraunhofer) region begins at

$$r_{ff} = \frac{2D^2}{\lambda_0}.$$

With $D = W = 67$ mm, $r_{ff} \approx 2(0.067)^2/0.1228 \approx 7.31$ cm. So any cable-length separation (typically ≥ 30 cm with the lab's blue SMAs) is comfortably in the far-field, and the azimuth pattern we measure will be the true radiation pattern rather than near-field coupling.

3 VNA Setup

The VNA was configured per the lab procedure with a 4-way quadrant split: top-left S_{11} Log-Mag, top-right S_{11} Smith (R+jX), bottom-left S_{22} Log-Mag, bottom-right S_{22} Smith (R+jX). Start frequency 1.5 GHz, stop frequency 2.7 GHz. The eCal module was used for calibration on both ports through the blue SMA cables. After the resonance was found in the WiFi band, the VNA was recalibrated over the narrower 2.35–2.55 GHz span.

¹<https://www.digikey.com/en/resources/conversion-calculators/conversion-calculator-pcb-trace-impedance>

4 Build the Antennas

Both antennas were built on single-sided copper-clad FR4. The patch and $50\ \Omega$ transmission line were cut from copper tape using an Xacto knife, with the transmission line meeting the patch at the center of the W -edge and running to the board edge where the side-launch SMA connector was soldered. The SMA center pin attaches to the transmission line and the body pins to the ground plane on the bare side of the board.

Photographs

Insert photo of Bao's antenna with all five dimensions labeled in Sharpie on the board itself: L , W , R , transmission line width, and transmission line length.

Insert photo of Nick's antenna with all five dimensions labeled in Sharpie on the board itself: L , W , R , transmission line width, and transmission line length.

As-built dimensions

	Bao's antenna (Y1)	Nick's antenna (Y2)
L initial cut (mm)	30.0	30.0
L after trim (mm)	26.0	26.7
W (mm)	67.0	67.0
Transmission line width (mm)	2.632	2.632
Transmission line length (mm)	<i>TBD</i>	<i>TBD</i>
Inset distance R (mm)	<i>TBD</i>	<i>TBD</i>

5 Initial Measurements and Impedance Matching

Initial resonance and trimming

With both antennas connected, S_{11} (Bao) and S_{22} (Nick) were swept from 1.5–2.7 GHz to find the initial resonance. The frequency with the lowest return-loss magnitude was treated as the operating frequency.

	Bao (Y1, S_{11})	Nick (Y2, S_{22})
Initial L (mm)	30.0	30.0
Initial f_{ant} (GHz)	<i>TBD</i>	<i>TBD</i>
Trim direction	cut down	cut down
Amount trimmed (mm)	4.0	3.3
Post-trim L (mm)	26.0	26.7
Post-trim f_{ant} (GHz)	≈ 2.443	≈ 2.443

Both antennas started at $L = 30$ mm but resonated well below 2.443 GHz on the first sweep. About 4 mm was shaved off Bao's antenna and 3.3 mm off Nick's to bring each resonance up into the WiFi

band. The 13% trim on Bao's patch is larger than the simple $L = c/(2f_c\sqrt{\epsilon_r})$ formula predicts, which lines up with the well-known fringing-field effect: the radiating aperture extends about $0.5h$ past each end of the metal patch, so the electrical length is roughly $L + 2(0.5h) \approx L + 1.4$ mm longer than the physical length. The formula treats only the geometric L , so the as-cut patches always resonate a bit low and we trim from there.

Input impedance measurement

At the post-trim operating frequency, the input impedance was read from the Smith chart marker.

	Bao's antenna	Nick's antenna
Z_{IN_0} from Smith chart (Ω)	<i>TBD</i>	<i>TBD</i>
Return loss at f_c (dB)	<i>TBD</i>	<i>TBD</i>

Inset feed calculation R

Moving the feed inward by distance R scales the input impedance as

$$Z_{IN_R} = Z_{IN_0} \cos^2\left(\frac{\pi R}{L}\right) \implies R = \frac{L}{\pi} \cos^{-1} \sqrt{\frac{Z_{IN_R}}{Z_{IN_0}}}$$

where $Z_{IN_R} = 50 \Omega$ is the target and Z_{IN_0} is the measured edge-fed impedance.

Calculated R (plug in measured L and Z_{IN_0}):

- Bao's antenna: $L = \textit{TBD}$, $Z_{IN_0} = \textit{TBD} \Omega \Rightarrow R = \textit{TBD mm}$
- Nick's antenna: $L = \textit{TBD}$, $Z_{IN_0} = \textit{TBD} \Omega \Rightarrow R = \textit{TBD mm}$

After inset cuts

Per the lab procedure, about half of the calculated R was cut from each side of the patch, then the VNA was monitored in Log-Mag mode and additional symmetric cuts made until return loss bottomed out.

	Bao's antenna (S_{11})	Nick's antenna (S_{22})
Final R (mm)	<i>TBD</i>	<i>TBD</i>
Return loss at f_c (dB)	<i>TBD</i>	<i>TBD</i>
Return loss at 2.400 GHz (dB)	<i>TBD</i>	<i>TBD</i>
Return loss at 2.487 GHz (dB)	<i>TBD</i>	<i>TBD</i>

Target: return loss below -20 dB at f_c , and at least -9.5 dB across the whole 2.400–2.487 GHz band.

VNA screen captures

Insert VNA screen capture (deliverable #1): both antennas connected, four-quadrant view of S_{11}/S_{22} Log-Mag +

6 More Measurements and Polarization Observations

Azimuth pattern

The Smith chart quadrants were reconfigured to show S_{12} and S_{21} Log-Mag instead. The two antennas were placed facing each other, separated by a fixed distance set by the VNA cable lengths. One antenna was held stationary; the other was rotated in 15° increments out to 90° . The aligned (0°) measurement was taken as the 0 dB baseline, and the values below are the drop from that baseline.

Angle ($^\circ$)	S_{21} (dB)	S_{12} (dB)	Drop from 0 dB baseline (dB)
0	TBD	TBD	0.0
15	TBD	TBD	TBD
30	TBD	TBD	TBD
45	TBD	TBD	TBD
60	TBD	TBD	TBD
75	TBD	TBD	TBD
90	TBD	TBD	TBD

Antenna separation distance: TBD cm .

Insert polar (azimuth) plot from Excel showing the radiation pattern across $\pm 90^\circ$. Use the 6+ data points above;

Cross-polarization test

With both antennas facing each other, one was tipped on its side so that its E -field axis was rotated 90° relative to the other. S_{21}/S_{12} was recorded, then compared to the value with the two antennas facing entirely away from each other.

Configuration	S_{21} (dB)	S_{12} (dB)
Both facing, one tipped 90° (cross-pol)	TBD	TBD
Both facing away from each other (back-to-back)	TBD	TBD

Discuss: is cross-pol lower than back-to-back, or higher? If lower, the patches are well-polarized and back-to-back

For the chosen separation, the far-field boundary is $r_{ff} = 2D^2/\lambda_0 \approx 7.31$ cm (using $D = W = 67$ mm and $\lambda_0 = 12.28$ cm at 2.443 GHz, derived above). Our antenna spacing of TBD cm is well past this threshold, so the link is operating in the far-field and the cross-pol result reflects true polarization mismatch rather than near-field coupling.

7 Discussion and Conclusion

Summarize: did the resonance land where the design calculation predicted (within a few percent)? How many tri